25901917 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCPHY200 - PRINCIPLES OF MECHANICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Under what condition is Newton's second law $F = ma$ applicable? [K] / [CO1]
2. A particle of mass $2\ kg$, initially at rest, is subjected to a force $F(t) = 2\ e^{2t}$. Determine $v(t)$ at $t = 2\ s$. [A] / [CO1]
3. If the air resistance force acting on an object is $F = -0.1\ v^2\ N$, what is the force when $v = 10\ m/s$? [A] / [CO1]
4. What is the difference between gravitational field intensity and gravitational potential? [U] / [CO2]
5. What is Newton's law of gravitation? Write the formula. [U] / [CO2]
6. How can we determine the magnitude and direction of force from the field lines representation of gravitational field? [U] / [CO2]
7. At what displacements is the potential energy of the simple harmonic oscillator is minimum and maximum? Explain . [U] / [CO3]
8. Plot the displacement-time graph for an overdamped harmonic oscillator. [A] / [CO4]

9. What is mean by a rigid body? [K] / [CO6]

10. Apply the concept of fictitious force to explain why passengers feel pushed outward in a turning car. [A] / [CO6]

11. Why does the day length on Earth slowly increase over millions of years? [U] / [CO6]

12. How can angular momentum of a system of particles be split into angular momentum of centre of mass and angular momentum about centre of mass. [An] / [CO6]

13. How does Kepler's third law show that planets farther from the Sun take longer to revolve around it.? [K] / [CO7]

14. What is effective potential energy in central force motion? Illustrate it with energy diagram for an attractive inverse square law field. [U] / [CO7]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15. Determine the motion of a particle if the force is a function of position of the particle [An] / [CO1]

16. Write a note on gravitational potential energy [U] / [CO2]

17. Derive the time period of a simple pendulum, and explain why it is valid only for small oscillations. [A] / [CO3]

18. Write the differential equation and general solution of a damped harmonic oscillator and discuss the case of an underdamped oscillator. [A] / [CO4]

19. State the differential equation of a forced damped oscillator and obtain the solution. [A] / [CO5]

20. What is mean by coriolis force? A gun fires a bullet horizontally at speed v from a place on the North pole of Earth. In which direction will it deflect due to coriolis force ? [A] / [CO6]

21. A particle of mass m has an initial velocity $v_0\ \hat{i}$. It is subjected to a constant force $F = f_0\ \hat{j}$. If L is the angiular momentum of the particle then evaluate $\frac{d^2L}{dt^2}$ at the initial time. [U] / [CO6]

22. Obtain the equations of motion for a planet moving around Sun in polar coordinates. [U] / [CO7]

[5x6 = 30]